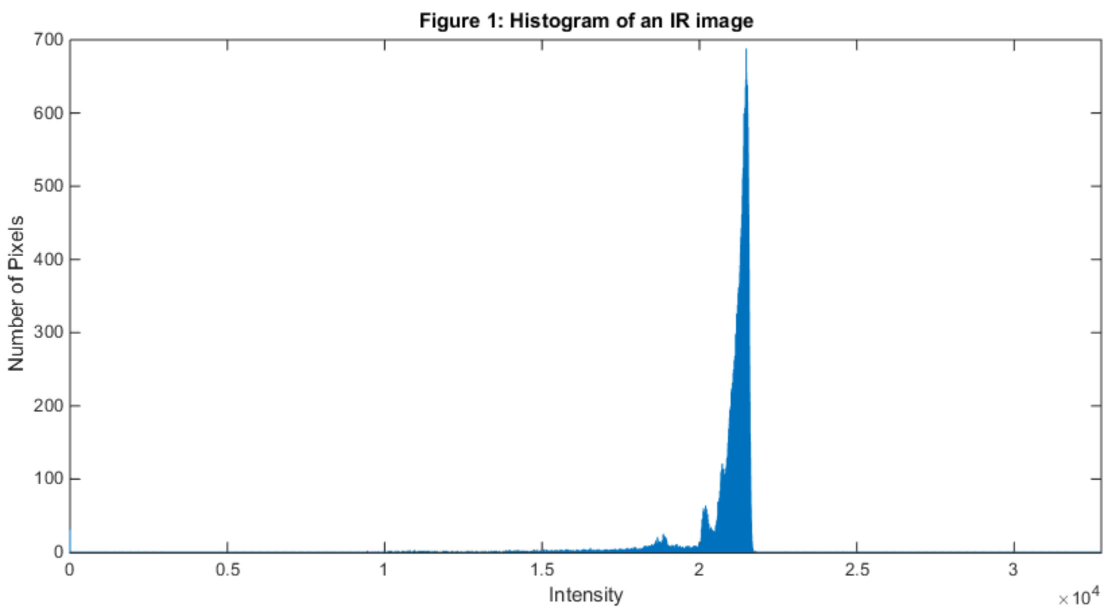
Histogram Based Automatic & Manual Gain Control

The purpose of this document is to describe the behavior and mathematical operations that goes in the Histogram Based Automatic and manual gain control. It is a part of the IRPU FW.

1. INTRODUCTION

The IR input image is usually grey with hardly any details showing. This is due to image intensity levels being concentrated in a tight area as shown in figure 1. This algorithm is used to stretch the input image range of intensity to the full range.



The histogram specify at each intensity level how many pixels have this intensity. The algorithm uses the histogram to stretch the intensity levels hence it was named histogram based. In the end it gives a table such as the one below.

Intensity Level	Number of pixels
0	0
1	4
2	117
3	55
…	
L-1	64
L	24

- L is the maximum intensity. i.e. 16383 for 14-bits.

The histogram usefulness is based on its ability to show at which intensity levels the pixels are concentrated. This gives an indication of the current intensity range of the image and allows through the usage of cumulative histogram to exclude a certain number of pixels from the calculations.

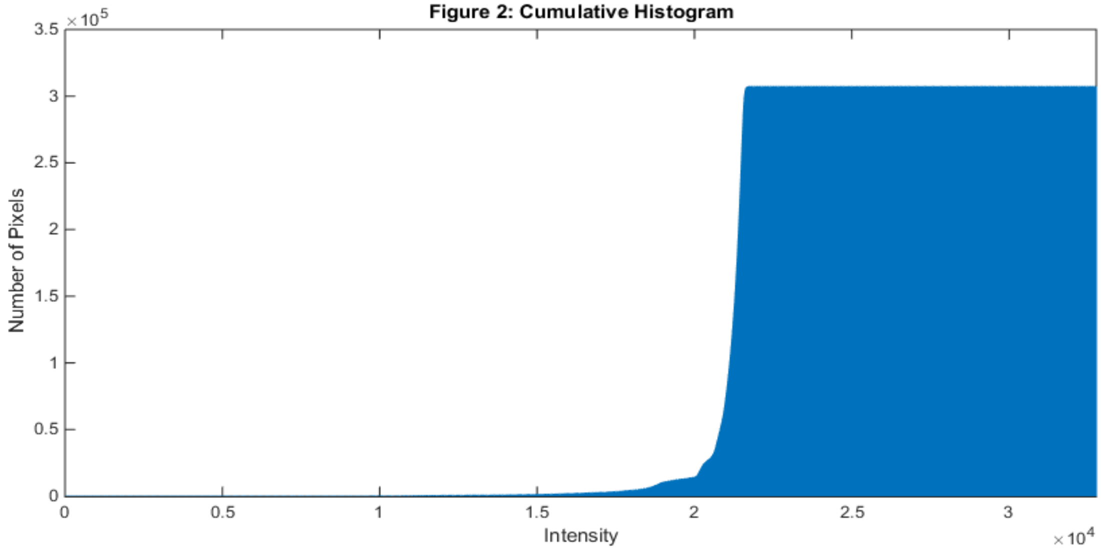
The cumulative histogram is derived from the histogram. It is basically the cumulative sum of the histogram. In further words, the number of pixels in each intensity level will also be combined with the number of pixels in all previous intensity levels. So, in the last intensity (i.e. 16384 for 14-bits) the value should equal the total number of pixels in the image or frame. It is used to determine at which intensity level a certain number of pixels is reached cumulatively. The below table draws comparison between the normal histogram and the cumulative histogram.

Intensity Level	Number of pixels	Number of pixels (cumulative)
0	0	0
1	4	4
2	117	121
3	55	176
4	118	294
5	212	506
…	…	…
L-1	64	307176
L	24	307200 (for 480x640 resolution)

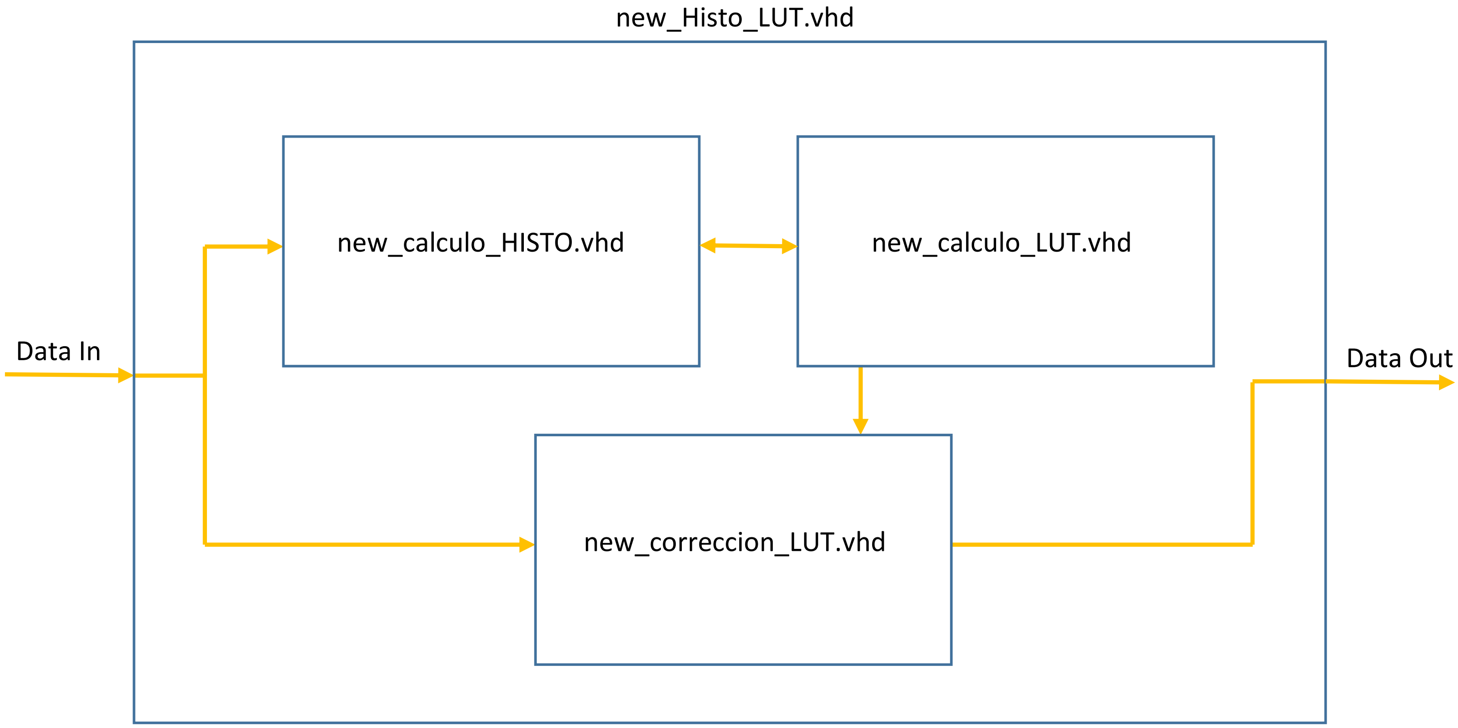
For example, if we have an image with a total pixels of 10000 (N) and we want to know at which intensity level we reach 1000 pixel, we would first calculate the histogram. Next, we calculate the cumulative histogram. Then, we would search in the cumulative histogram table for the first intensity with a value equal to 1000 or higher. This level is what we are looking for.

Figure 2 is the cumulative histogram for the image in figure 1.

Figure 2: Cumulative Histogram



BLOCK DIAGRAM



The main module consists of three sub modules:

1. new_calculo_HISTO.vhd

This sub module receives the input data and calculate its histogram. After calculation, it stores the histogram in an internal Look Up Table (LUT).

2. new_correccion_LUT.vhd

This sub module receives the input data, insert it as an address in its internal LUT and output the corresponding values. It does not perform any kind of calculations.

3. new_calculo_LUT.vhd

This is the main calculation module. It obtains the histogram from new_calculo_HISTO.vhd and calculates the cumulative histogram and then store it in the same LUT. After that, it calculates the conversion LUT table and store it in the internal LUT inside new_correccion_LUT.vhd

ALGORITHM DESCRIPTION

The design include four options selected by the software signal tipo_calculo_lut_Micro_Ext. They are as follows:

1. Linear
2. Auto Brightness
3. Manual
4. Non-Linear

A detailed explanation will be given on each method.

1. Linear

The concept behind the linear method is the stretching concept. Since we have an image whose intensity is very compact, we want to stretch this intensity to the full output range. But then, some outliner pixels may affect all the data by being far from the rest of the pixels. To solve this problem, a small percentage of outliner pixels is ignored when making calculations. There exist four different percentages to choose from based on the value of the software control signal M_ancho_vent_lineal:

a. 1/64 (1.56%)
b. 1/128 (0.78%)
c. 1/256 (0.39%)
d. 1/512 (0.20%).

For example, if the total number of Pixels (N) = 10000, and we want to ignore 1% of the data as outliner pixels. Note that this percentage is only for one side and we have two side: inferior and superior. So in total, 2% of the data will be ignored.

$$Outliner\_low = 10000 \times 0.01 = 100$$

This means the lowest 100 pixels in the image will be ignored. The same is done to highest 100 pixels as well.

$$Outliner\_up = N - Outliner\_low = 10000 - 100 = 9900$$

By using the cumulative histogram, the intensity in which the Outliner_low is reached is the inferior limit. And the intensity in which the Outliner_up is reached is the superior limit. By subtracting inferior limit from superior limit, the range of the considered input data (Window) will be obtained.

$$Window = superior\ limit - inferior\ limit$$

The data in this window will be stretched to cover the entire output range.

After calculating the window, the gain factor will be calculated, this is the factor that will be multiplied with the intensities to change the range to 255 or whatever range desired.

$$Gain\ Factor = \frac{255}{Window}$$

Now we consider the values in the window which are higher than the inferior limit and lower than the superior limit. A counter (C) will be initialized starting from the inferior limit and for every intensity it will increase by one. The value of the counter each time is multiplied by the Gain factor and stored as the corresponding new value for that intensity. This goes on until the superior limit is reached. By the end, a LUT that looks like the one below will be gained.

Consider the Inferior limit = 10000, superior limit = 12000, Window = 2000,

$$Gain\ Factor = \frac{255}{Window} = \frac{255}{2000} = 0.1275$$

Address	Data Out	Notes
0	0	Values lower than inferior limit are assigned 0.
1	0	
2	0	
…	…	
10000	0 * 0.1275 = 0	Inferior limit
10001	1 * 0.1275 = 0.1275	The initialized counter (C) will be multiplied by the gain factor.
10002	2 * 0.1275 = 0.255	
…	…	
11998	1998 * 0.1275 = 254.745	
11999	1999 * 0.1275 = 254.8725	
12000	2000 * 0.1275 = 255	Superior limit
12001	255	Values higher than superior limit are assigned 255.
12002	255	
…	…	
32767	255	

Now when the image or the frame arrives, each pixel will be inserted in the LUT and according to its intensity value a corresponding output will be given.

Although the algorithm is dynamic, it gives the user some control over the inferior and the superior limit through the software part with the signals M_ensanche_vent_lineal_up & M_ensanche_vent_lineal_dw. Based on control values the inferior limit could be reduced by a certain amount. The same thing also exist for the superior limit, but it will be increased instead of being reduced.

2 & 3. Auto Brightness & Manual

Those two methods were grouped together because they are nearly identical. They are both considered a manual control. When those methods are selected, the software first selects the linear method. The reason for that is to use the inferior limit and the superior limit calculated by the linear method as default starting values. The difference between the two methods is in their manual override to inferior and superior limits.

In Manual method, when increasing the gain, the window size is reduced and the accordingly the inferior limit is increased to compensate for the reduced size of the window. Superior limit is always calculated by adding Inferior limit and window size to each other. When decreasing the gain, the window size is increased and the inferior limit is decreased to accommodate the increased window.

$$Superior\ limit = Inferior\ limit + Window$$

In Auto Brightness method, the inferior limit does not change. So the change is always in the superior limit and window size. When increasing the gain, the window size is reduced and accordingly the superior limit is deceased while the inferior limit remains unchanged. When decreasing the gain, the window size is increased and the superior limit is increased while the inferior limit remains unchanged.

The offset control for both methods is identical. It controls the inferior limit. So, by increasing the offset, the inferior limit is increased. And by decreasing the offset, the inferior limit is decreased.

The rest of the two methods are completely identical to the linear method. They both calculate the gain factor and then the LUT in a similar way.

4. Non-Linear

The non-Linear method have a different concept from the other methods. It does not use superior and inferior limits. It uses logarithms to put more emphasis on the lower intensities. So, the lower intensities would appear clearer in the output.

It starts by fetching the histogram data, i.e. the number of pixels for each intensity (N). The number of pixels would be entered in a base 2 logarithmic table inside the module that produce the result as an integer (fractions are ignored). Since the maximum number that the logarithmic table would receive is the total number of pixels (T), a table with T number of entries would be needed. The signal that is able to contain that value is 19 bits wide.

$$(total\ number\ of\ pixels)\ T = 480 * 640 = 307200\ pixel$$

However, since there will be lot of redundancy in the table, a trick is used to reduce the size of the table from 307200 entries to only 1024 entries. The table accept only 10 bits signal as input (will refer to the signal as P). It will deal with the input in two ways according to its value:

1. The first case, if N < 512. In this case it will be assigned to the rightmost 8 bits. And everything will go normally.

$$P = '0'\ \&\ N(8\ downto\ 0)$$

2. The second case, if N ≥ 512. In this case the leftmost bit of P will be toggled on and the rightmost 8 bits of P will be assigned to the 17-9 bits of N.

$$P = '1'\ \&\ N(17\ downto\ 9)$$

The second half of the table is modified accordingly. The values are mapped to 512 value rather than 306688 (307200 – 512) values. This will not have any effect on the accuracy of the output since we take only the integer part. Hence the size of the table was reduced drastically without any loss of accuracy.

After that, those results will be returned to the histogram table accumulatively (i.e. each entry will equal the entry itself plus the previous entries. Then the gain factor will be calculated as follows:

$$Gain\ factor = \frac{255}{The\ last\ entry\ in\ the\ modified\ histrogam\ table}$$

By multiplying this gain factor by each of the table entries, the entries will be mapped from 0 to 255. This method will result in an output that put more emphasis on low intensities and show a better image in that regard than the linear method.